

Project 2: Gossip Protocol

Project group: 24

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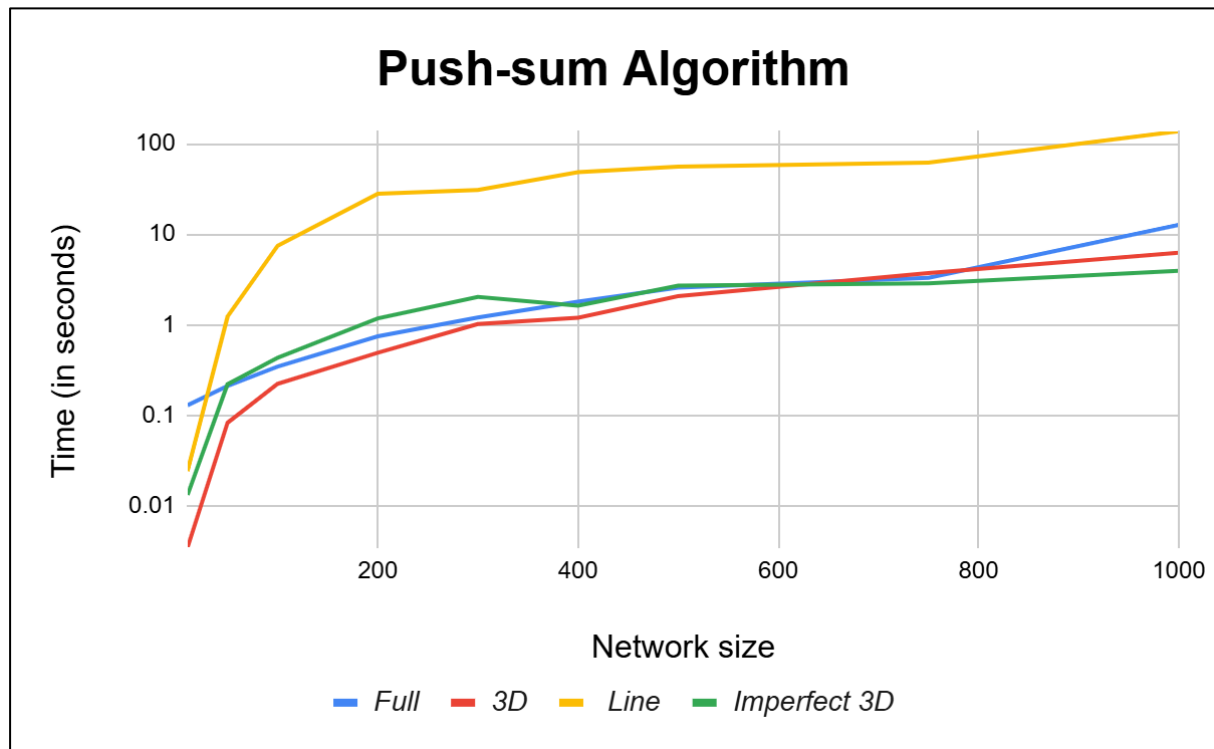
Results:

For both algorithms, we tested for network sizes ranging from 10 to 1000. We recorded the results and plotted line graphs for both algorithms. These line graphs, having the network size on the X axis and the convergence time on the Y axis, show the trend for all 4 topologies. A logarithmic scale was used on the Y axis to get a better understanding of the graph results.

Convergence time results for Push-sum algorithm:

Network size	Convergence time (in seconds)			
	Full Network	3D Grid	Line	Imperfect 3D Grid
10	0.129086397	0.0034816	0.0241664	0.0132096
50	0.211204797	0.0827392	1.2332032	0.2200576
100	0.345082171	0.22272	7.4679296	0.4322304
200	0.745333074	0.4907008	27.9830528	1.1756544
300	1.201180099	1.0167296	30.7459072	2.0287488
400	1.795578644	1.1920384	48.4753408	1.6243712
500	2.569007177	2.073088	55.6645376	2.7024384
750	3.305522391	3.719168	61.5245312	2.8709888
1000	12.65877919	6.2260224	136.7286784	3.943936

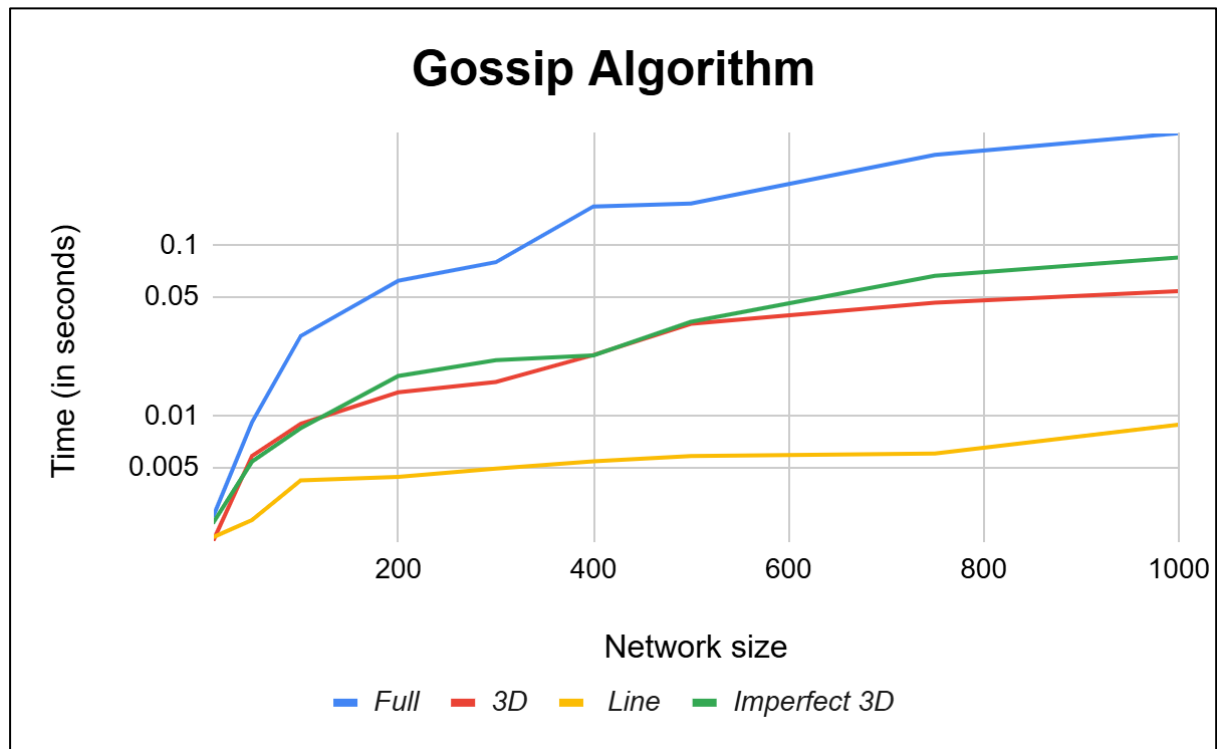
Line graph for Push-sum algorithm:



Convergence time results for Gossip algorithm:

Network size	Convergence time (in seconds)			
	Full Network	3D Grid	Line	Imperfect 3D Grid
10	0.00256	0.0018432	0.0019456	0.0023552
50	0.009216	0.0058368	0.0024576	0.0054272
100	0.0294912	0.0090112	0.0041984	0.0084992
200	0.0623616	0.013824	0.0044032	0.0172032
300	0.0800768	0.015872	0.0049152	0.0212992
400	0.1700864	0.0228352	0.0054272	0.0227328
500	0.1768448	0.034816	0.0058368	0.03584
750	0.3418112	0.0462848	0.0060416	0.06656
1000	0.4566016	0.0540672	0.0089088	0.0852992

Line graph for Gossip algorithm:



Interesting findings:

- **Topology and Convergence Speed:** The nature of the graphs for both algorithms conveys the rate at which convergence is achieved for a topology. For example, the **Line topology** tends to take the longest for the **Push-Sum** algorithm to converge, however, the same topology converges faster for the **Gossip** algorithm.
- **Graph Curvature and Convergence Time:** The curvature of the graphs observed in the simulations follows a pattern similar to logarithmic curves. This suggests that the approximate convergence time for both **Gossip** and **Push-Sum** algorithms scales with $O(\log N)$.
- **S/W Ratio:** The ratio of the sum of actors (S) to the sum of weights (W) approaches an approximate formula of

$$\frac{N * (N + 1)}{2 * N} = \frac{(N + 1)}{2}$$

which represents the **average** of all node values, indicating each node has the average value propagated to it.